

Multiple Choice (10 marks)

1. Most animal cells, regardless of species, are relatively small and about the same size. Relative to larger cells, why is this?
 - (a) Smaller cells are more resilient to external damage.
 - (b) Smaller cells have a larger number of organelles.
 - (c) Smaller cells require less energy to function.
 - (d) Smaller cells can multiply faster.
 - (e) Smaller cells fit together more tightly.
2. The average birth weight for human babies is about 3 kg. Which of the following terms best describes this?
 - (a) disruptive selection
 - (b) sexual selection
 - (c) bottleneck effect
 - (d) non-random mating
 - (e) stabilizing selection
3. Supposing that the clonal selection theory is correct, how can an organism genetically make such diverse proteins?
 - (a) Through vertical gene transfer from parent to offspring
 - (b) Through random assortment of genes during meiosis
 - (c) Through peptide bond formation errors during translation
 - (d) Through genetic mutation only
 - (e) Through transcriptional errors in RNA processing
4. Using conditioning, how can one determine the sensitivity of a dog's hearing to sounds of different frequencies?
 - (a) By using a controlled group of dogs not exposed to the sounds
 - (b) By using sensory deprivation to heighten hearing sensitivity
 - (c) By observing the dog's natural behavior in different sound environments
 - (d) By using negative reinforcement only
 - (e) By using operant conditioning
5. If a random variable X is normally distributed with a mean of 118 and a standard deviation of 11, what Z-scores correspond to raw scores of 115, 134, and 99?

- (a) -.27, 1.45, 1.73
- (b) -.27, -1.45, -1.73
- (c) -2.18, 0.15, -1.82
- (d) 0, 1.5, -2
- (e) .27, 1.45, 1.73

True / False (10 marks)

Answer True or False

6. True or False: Chemical shift MRI: is there any contribution to morphologic evaluation of solid breast masses?
7. True or False: Motor performance in chronic low back pain: is there an influence of pain-related cognitions?
8. True or False: Does the National Institutes of Health Stroke Scale favor left hemisphere strokes?
9. True or False: Should prostate specific antigen be adjusted for body mass index?
10. True or False: Does a colonoscopy after acute diverticulitis affect its management?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the importance of interdisciplinary collaboration in understanding the biosphere, focusing on the roles of climatologists, ecologists, and geologists in mapping climatic and vegetation patterns.
12. Discuss the mechanisms that ensure the unidirectional nature of chemical transmission at synapses, focusing on the role of neurotransmitter localization and synaptic structure.

End of Paper